

PROJECT REPORT

EXPERIMENT - 12

Project Statement

Design and develop an Industrial Diesel Generator Monitoring and Maintenance System. The system monitors generator performance parameters and supports preventive maintenance operations

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Industrial Diesel Generator Monitoring and Maintenance System

Project Overview

The Industrial Diesel Generator Monitoring and Maintenance System is an IoT-based system designed to monitor important operating parameters of a diesel generator and support preventive maintenance activities. The system uses an ESP8266 Node MCU as the main controller, an ACS712 current sensor to monitor generator load current, and a DHT11 sensor to monitor temperature and humidity around the generator.

The collected sensor data is processed by the ESP8266 to identify abnormal operating conditions. A buzzer provides an immediate warning when the monitored parameters exceed predefined safe limits. This system can help operators detect overheating, excessive electrical load, and other abnormal conditions at an early stage, thereby reducing the possibility of unexpected generator shutdown and supporting timely maintenance.

Main Objectives:

- To develop an IoT-based diesel generator monitoring system.
- To continuously monitor generator current consumption.
- To monitor temperature and humidity conditions near the generator.
- To detect abnormal operating conditions using predefined threshold values.
- To provide an audible warning through a buzzer during abnormal conditions.
- To support preventive maintenance by identifying potential problems early.
- To reduce unexpected generator downtime.
- To improve generator operational safety and reliability.

Problem Statement

Industrial diesel generators are commonly used as backup and continuous power sources. Continuous operation under excessive electrical load or unfavorable environmental conditions can affect generator performance and reliability. Manual monitoring may not identify abnormal conditions quickly enough, resulting in overheating, excessive loading, equipment damage, or unexpected shutdown.

Therefore, an Industrial Diesel Generator Monitoring and Maintenance System is proposed to continuously monitor important generator parameters and provide warning indications when abnormal conditions are detected. The system supports preventive maintenance by allowing operators to identify potential problems before they develop into major failures.

Problems Addressed:

The proposed system addresses the following problems:

- Lack of continuous generator parameter monitoring.
- Difficulty in identifying excessive electrical load.
- Delayed detection of high-temperature conditions.
- Dependence on manual inspection.
- Unexpected generator shutdown.
- Increased possibility of equipment damage.
- Difficulty in performing timely preventive maintenance.
- Reduced generator reliability due to undetected abnormal conditions.

Proposed Solution

The proposed system uses an ESP8266 Node MCU to collect and process sensor information. The ACS712 current sensor measures the electrical current associated with the generator load, while the DHT11 sensor monitors temperature and humidity.

The ESP8266 compares the measured values with predefined threshold levels. Under normal conditions, the system continues monitoring. If an abnormal current or temperature condition is detected, the ESP8266 activates the buzzer to alert the operator.

The monitoring information can also be extended to an IoT/cloud platform through the ESP8266 Wi-Fi capability for remote observation and maintenance analysis.

Proposed Operation:

The system operates in the following sequence:

- The ESP8266 Node MCU is powered on.
- The ACS712 current sensor measures the generator load current.
- The DHT11 sensor measures temperature and humidity.
- Sensor readings are received by the ESP8266.
- The controller processes and compares the readings with predefined limits.
- If the parameters remain within acceptable limits, normal monitoring continues.
- If an abnormal condition is detected, the buzzer is activated.
- The operator can inspect the generator and perform suitable preventive maintenance.
- The monitoring cycle repeats continuously.

System Architecture

The Industrial Diesel Generator Monitoring and Maintenance System is designed using a layered IoT architecture consisting of an Input Section, Processing Section, Communication Section, Cloud Section, and User Section. The system continuously collects generator operating parameters, processes the sensor data, identifies abnormal conditions, and provides alerts to support preventive maintenance.

Input Section:

The Input Section consists of sensors used to collect important generator and environmental parameters.

- **ACS712 Current Sensor:** Measures the electrical current drawn by the generator load.
- **DHT11 Sensor:** Measures the temperature and humidity around the generator.
- **Buzzer:** Acts as an output alert device when abnormal conditions are detected.

The sensor readings are continuously supplied to the ESP8266 Node MCU for processing.

Processing Section:

The ESP8266 Node MCU acts as the main processing unit of the system.

Its functions include:

- Reading current values from the ACS712 sensor.
- Reading temperature and humidity values from the DHT11 sensor.
- Processing and filtering the sensor readings.
- Comparing measured values with predefined threshold limits.
- Determining the generator operating condition.
- Activating the buzzer when an abnormal condition is detected.
- Preparing sensor data for IoT/cloud transmission.

The system can classify the generator condition as Normal, Warning, or Maintenance Required depending on the monitored parameters.

Communication Section:

The ESP8266 Node MCU provides Wi-Fi communication for transmitting the collected generator data.

The communication section can:

- Send sensor readings through Wi-Fi.
- Transfer data to an IoT/cloud platform.
- Support remote generator monitoring.
- Provide periodic updates of generator operating conditions.
- Enable future mobile or web-based monitoring.

For the basic prototype, the Wi-Fi communication feature can be demonstrated using a suitable IoT platform or local network.

Cloud Section:

The Cloud Section is an optional extension for storing and analyzing generator monitoring data.

The cloud platform can be used to:

- Store current, temperature, and humidity readings.
- Display real-time generator parameters.
- Maintain historical data.
- Generate graphs and trends.
- Identify repeated abnormal conditions.
- Support preventive maintenance analysis.
- Provide remote access to monitoring information.

In an advanced version, historical data can be analyzed to identify patterns that may indicate developing generator problems.

User Section:

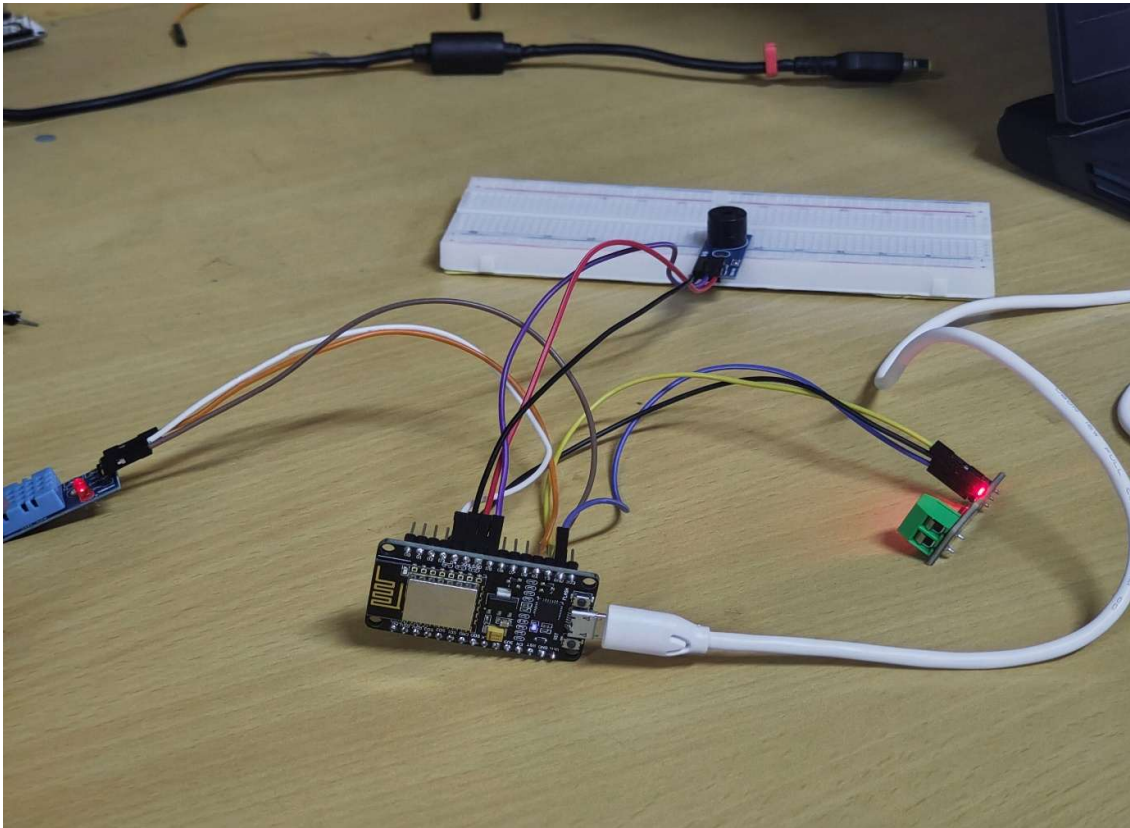
The User Section provides the generator operator or maintenance personnel with information about the generator's condition.

The user can:

- Observe generator parameters.
- Receive abnormal-condition alerts.
- Identify excessive current conditions.
- Monitor temperature and humidity.
- Inspect the generator when an alarm occurs.
- Schedule preventive maintenance.
- Review historical operating data when cloud monitoring is implemented.

The buzzer provides an immediate local alert, while a future IoT implementation can provide remote alerts through a web or mobile interface.

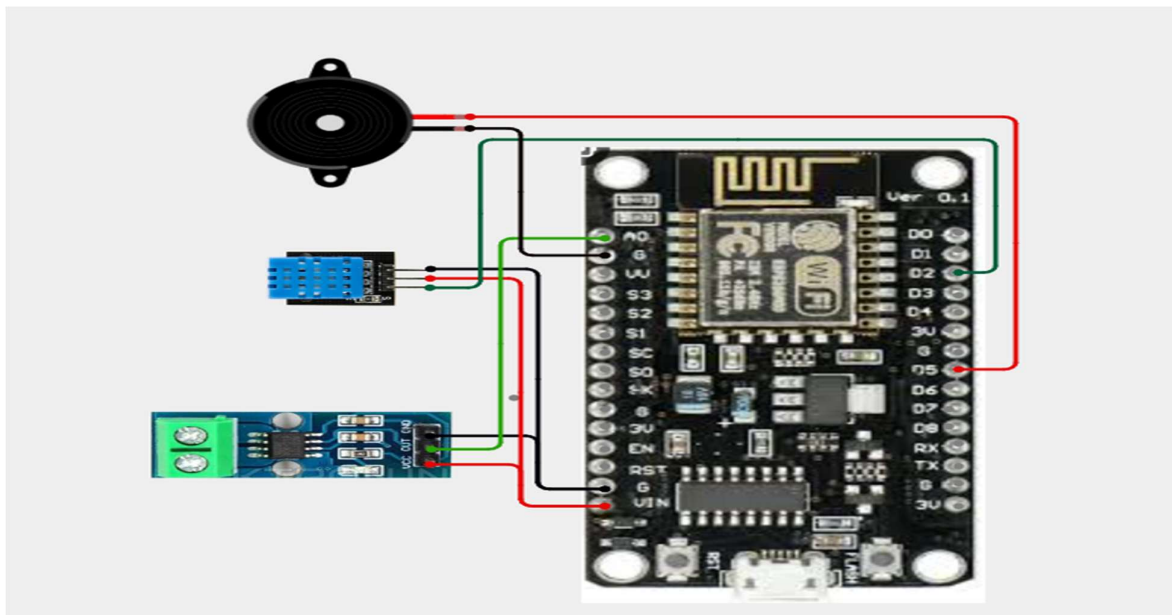
Image:



Circuit Table

S. No	Component	Pin/Terminal	ESP8266 Node MCU Pin	Function
1	ACS712 Current Sensor	VCC	3.3V / suitable supply	Provides power to current sensor
2	ACS712 Current Sensor	GND	GND	Common ground
3	ACS712 Current Sensor	OUT	A0	Sends analog current signal to ESP8266
4	DHT11 Sensor	VCC	3.3V	Provides power to DHT11
5	DHT11 Sensor	GND	GND	Common ground
6	DHT11 Sensor	DATA	D4 (GPIO 2)	Sends temperature and humidity data
7	Buzzer	Positive (+)	D5 (GPIO14)	Generates alarm during abnormal conditions
8	Buzzer	Negative (-)	GND	Completes buzzer circuit
9	ESP8266 Node MCU	USB	USB Power	Provides power and programming interface

Circuit Design



Important Pin Mapping:

- **ESP8266 A0** → ACS712 OUT
- **ESP8266 D4 (GPIO2)** → DHT11 DATA
- **ESP8266 D5 (GPIO14)** → Buzzer positive terminal
- **ESP8266 3.3 V** → DHT11 VCC
- **ESP8266 GND** → DHT11 GND
- **ESP8266 GND** → ACS712 GND
- **ESP8266 GND** → Buzzer negative terminal

Algorithm

Step 1: Start the system.

Step 2: Initialize the ESP8266, ACS712 sensor, DHT11 sensor, and buzzer.

Step 3: Read the current value from the ACS712 sensor.

Step 4: Read temperature and humidity from the DHT11 sensor.

Step 5: Process and display/store the sensor readings.

Step 6: Compare the current reading with the predefined current limit.

Step 7: Compare the temperature reading with the predefined temperature limit.

Step 8: If any parameter exceeds its threshold, activate the buzzer.

Step 9: If all parameters are within the acceptable range, keep the buzzer OFF.

Step 10: Repeat the monitoring process continuously.

Coding

```
#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

#include <DHT.h>


#define DHTPIN 2

#define DHTTYPE DHT11


#define CURRENT_PIN A0

#define BUZZER 5


const char* ssid = "Rithick";

const char* password = "12345678";


unsigned long channelID = 3462650;

const char* writeAPIKey = "IMFMH9GRDZ1QUFQB";


WiFiClient client;


DHT dht(DHTPIN, DHTTYPE);


void setup()
{
    Serial.begin(9600);


    pinMode(BUZZER, OUTPUT);
```

```
digitalWrite(BUZZER, LOW);
```

```
dht.begin();
```

```
WiFi.begin(ssid, password);
```

```
Serial.print("Connecting");
```

```
while (WiFi.status() != WL_CONNECTED)
```

```
{
```

```
    delay(500);
```

```
    Serial.print(".");
```

```
}
```

```
Serial.println();
```

```
Serial.println("WiFi Connected");
```

```
ThingSpeak.begin(client);
```

```
}
```

```
void loop()
```

```
{
```

```
    float temperature = dht.readTemperature();
```

```
    float humidity = dht.readHumidity();
```

```
    if (isnan(temperature) || isnan(humidity))
```

```
{  
  Serial.println("DHT Error");  
  delay(2000);  
  return;  
}  
  
int sensorValue = analogRead(CURRENT_PIN);  
  
float voltage = sensorValue * (3.3 / 1023.0);  
  
float current = (voltage - 2.5) / 0.185;  
  
if(current < 0)  
  current = -current;  
  
Serial.println("-----");  
Serial.print("Temperature : ");  
Serial.println(temperature);  
  
Serial.print("Humidity   : ");  
Serial.println(humidity);  
  
Serial.print("Current    : ");  
Serial.print(current);  
Serial.println(" A");
```

```
if(current > 3.0 || temperature > 40)
```

```
{
```

```
    digitalWrite(BUZZER, HIGH);
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(BUZZER, LOW);
```

```
}
```

```
ThingSpeak.setField(1, current);
```

```
ThingSpeak.setField(2, temperature);
```

```
ThingSpeak.setField(3, humidity);
```

```
int response = ThingSpeak.writeFields(channelID, writeAPIKey);
```

```
if(response == 200)
```

```
{
```

```
    Serial.println("Uploaded to ThingSpeak");
```

```
}
```

```
else
```

```
{
```

```
    Serial.print("Upload Failed. Error Code: ");
```

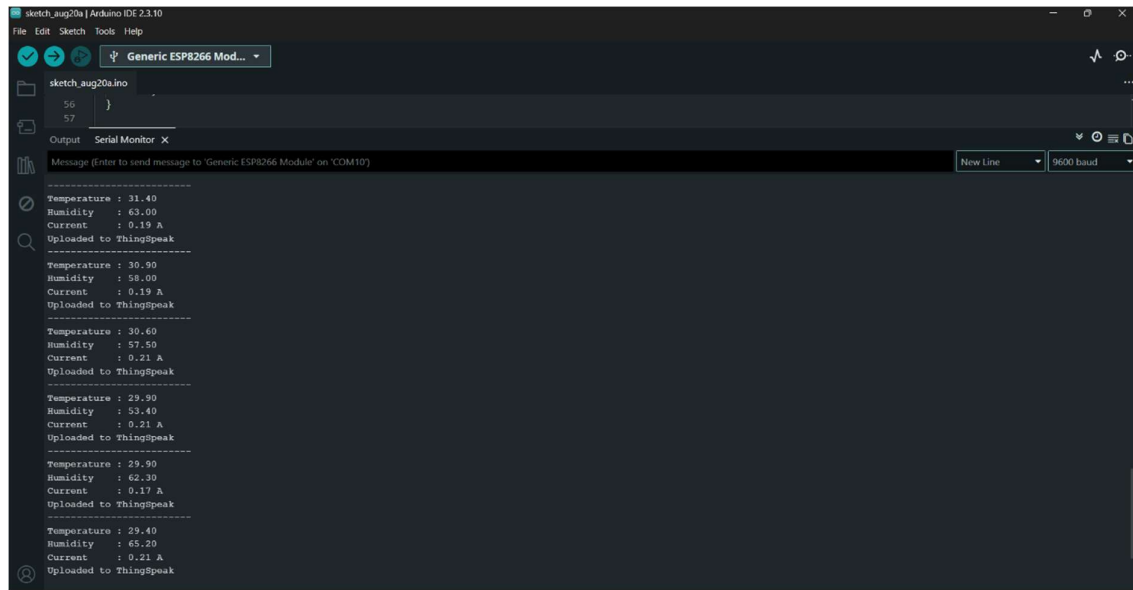
```
    Serial.println(response);
```

```
}
```

```
delay(20000);
```

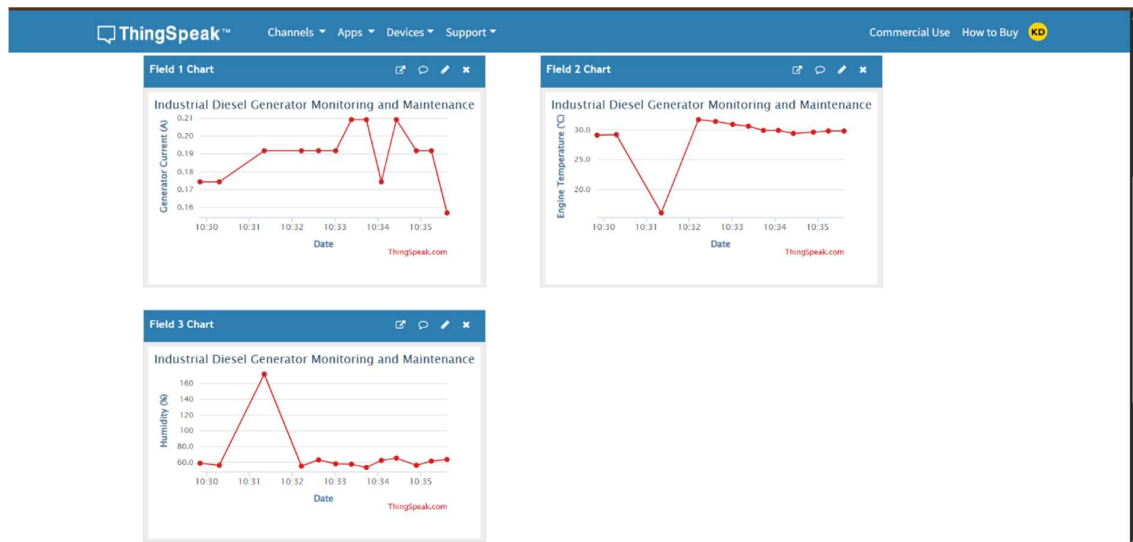
```
}
```

Coding Output:



The screenshot shows the Arduino IDE interface with the Serial Monitor open. The monitor displays a series of sensor readings being sent to ThingSpeak. The data includes Temperature, Humidity, and Current, each followed by the text 'Uploaded to ThingSpeak'.

```
sketch_aug20a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
sketch_aug20a.ino
56 }
57 }
Serial Monitor X
Output
Message (Enter to send message to 'Generic ESP8266 Module' on 'COM10')
New Line 9600 baud
Temperature : 31.40
Humidity : 63.00
Current : 0.19 A
Uploaded to ThingSpeak
Temperature : 30.90
Humidity : 58.00
Current : 0.19 A
Uploaded to ThingSpeak
Temperature : 30.60
Humidity : 57.50
Current : 0.21 A
Uploaded to ThingSpeak
Temperature : 29.90
Humidity : 53.40
Current : 0.21 A
Uploaded to ThingSpeak
Temperature : 29.90
Humidity : 62.30
Current : 0.17 A
Uploaded to ThingSpeak
Temperature : 29.40
Humidity : 65.20
Current : 0.21 A
Uploaded to ThingSpeak
```



System Working

During operation, the ACS712 sensor continuously monitors the electrical current drawn by the generator load. The DHT11 sensor simultaneously measures the surrounding temperature and humidity. These sensor outputs are sent to the ESP8266 Node MCU.

The ESP8266 processes the received values and checks whether they are within predefined operating limits. When the generator operates under normal conditions, the buzzer remains OFF. If excessive current or high temperature is detected, the controller recognizes the condition as abnormal and activates the buzzer.

This warning allows the operator to inspect the generator and take preventive action, such as reducing the connected load, checking ventilation, inspecting the generator, or scheduling maintenance.

Bill of Materials

S. No	Component	Quantity
1	ESP8266 Node MCU	1
2	ACS712 Current Sensor	1
3	DHT11 Temperature & Humidity Sensor	1
4	Buzzer	1
5	Breadboard	1
6	Jumper Wires	1 Set
7	USB Cable	1
8	5V/USB Power Supply	1

Prototype Implementation

The prototype is implemented on a breadboard using the ESP8266 Node MCU as the central controller. The ACS712 current sensor is connected to the analog input of the Node MCU, while the DHT11 sensor is connected to a digital GPIO pin. The buzzer is connected to another GPIO pin for generating an audible warning.

The sensors are positioned so that the ACS712 can measure the generator load current and the DHT11 can sense the temperature and humidity conditions around the generator. The ESP8266 is programmed to continuously collect sensor data and make decisions according to the defined threshold values.

Prototype Working

When the prototype is powered, the ESP8266 initializes all connected components. The ACS712 begins measuring current while the DHT11 measures temperature and humidity. The ESP8266 periodically reads these values and evaluates the generator's operating condition.

During normal operation, the system continues monitoring without activating the alarm. When the current or temperature exceeds the configured threshold, the buzzer turns ON, indicating that the generator requires attention. After the abnormal condition is corrected and the measured values return to the safe range, the buzzer can be switched OFF and normal monitoring resumes.

Expected Prototype Output

The prototype is expected to provide:

- Continuous generator current monitoring.
- Temperature measurement near the generator.
- Humidity measurement near the generator.
- Normal-condition monitoring.
- High-current warning.

- High-temperature warning.
- Audible buzzer indication during abnormal conditions.
- Support for preventive maintenance decisions.

Prototype Demonstration

During the demonstration, the system is first powered on and allowed to initialize the sensors. A normal generator-load condition is simulated, and the current, temperature, and humidity values are monitored. The buzzer remains OFF when all parameters are within their predefined limits.

An abnormal condition can then be simulated by increasing the monitored current or introducing a higher-temperature condition near the DHT11 sensor. When the measured value crosses the selected threshold, the ESP8266 activates the buzzer. This demonstrates the system's ability to detect abnormal operating conditions and provide an immediate maintenance warning.

Results and Performance Analysis

The developed Industrial Diesel Generator Monitoring and Maintenance System successfully demonstrates the monitoring of important generator operating parameters using the ESP8266 Node MCU. The ACS712 current sensor measures the electrical load current, while the DHT11 sensor monitors the temperature and humidity around the generator.

The ESP8266 processes the sensor readings and compares them with predefined threshold values. Under normal operating conditions, the system continues monitoring without activating the alarm. When an abnormal current or temperature condition is detected, the buzzer is activated to provide an immediate warning to the operator.

The prototype therefore demonstrates the following results:

- Continuous monitoring of generator load current.
- Temperature and humidity monitoring.

- Automatic identification of abnormal conditions.
- Immediate audible warning through the buzzer.
- Reduced dependence on manual monitoring.
- Support for preventive maintenance decisions.
- Potential for future cloud-based remote monitoring.

Performance Analysis:

The performance of the system can be evaluated based on sensor response, monitoring continuity, abnormal-condition detection, and alarm response.

Performance Parameter	Observation
Current Monitoring	ACS712 continuously monitors generator/load current
Temperature Monitoring	DHT11 provides periodic temperature readings
Humidity Monitoring	DHT11 provides environmental humidity readings
Abnormal Detection	ESP8266 compares readings with predefined thresholds
Alarm Response	Buzzer activates when an abnormal condition is detected
Monitoring Continuity	System can continuously repeat the sensing and analysis cycle
Maintenance Support	Abnormal readings provide an indication for inspection
IoT Expandability	ESP8266 Wi-Fi enables future cloud monitoring